MILITARY SPECIFICATION

ANTENNA AT-781A/U

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers one type of UHF Antenna, the AT-781A/U, a circularly polarized omnidirectional antenna with a frequency range of 406 through 549 megacycles (mc) for use aboard ship.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-C-17	Cables, Radio Frequency; Coaxial, Dual Coaxial, Twin Conductor, and Twin Lead.
MIL-C-71	Connectors, "N", for Radio Frequency Cables.
MIL-E-4158	Electronic Equipment, Ground: General Requirements for.
MIL-T-17113	Tests; Shock, Vibration, and Inclination (for Electronic Equipment) General Specification.
MIL-P-17555	Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts.
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment.

NAVY DEPARTMENT

General Specifications for Inspection of Material.

2.2 Availability of Documents:

(1) When requesting applicable documents give both the title and number. All requests shall be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, (AD), Philadelphia 11, Pennsylvania.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:52Z
Check the source to verify that this is the current version before use.

2.3 Precedence of Documents - When the requirements of the contract, this specification or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over all referenced documents.

(2) This Specification - This specification shall have precedence over all applicable subsidiary documents. Deviations from this specification, or from subsidiary documents where applicable, may be made only upon written approval.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts, Material and Processes - Material shall be in accordance with Specification MIL-E-4158, except as specified herein. In this regard, the selection of material shall be based on the fulfillment of the major design objectives as the prime consideration.

3.2 Design and Construction

3.2.1 General - The equipment shall meet the requirements of Specification MIL-E-4158, except as specified herein. The design objectives of a high-strength-to-weight ratio, reliability, simplicity, ease of installation and maintenance consistent with the need of the Military service, shall be considered during all phases of equipment design and construction.

3.2.2 Service Conditions

3.2.2.1 Temperature - The equipment shall not be affected in operation by ambient environmental temperatures of minus (-) 26 degrees to 125 degrees Fahrenheit (F) with additional heating of the sun.

3.2.2.2 Wind Load - The equipment shall be capable of withstanding a 120-knot wind.

3.2.2.3 Salt Atmosphere - The equipment shall be capable of operating without failure for a period normal to similar electronic equipment aboard ship, without being adversely affected by humid salt atmosphere.

3.2.2.4 Shock and Vibration - The complete antenna shall meet the applicable shock, vibration, and inclination requirements of Specification MIL-T-17113.

3.2.3 Construction - Equipment shall be constructed to conform to the service conditions as specified herein.

3.2.3.1 Mating Surfaces - Mating surfaces of dissimilar metals shall be protected to prevent corrosion according to paragraph 3.1.8.2 of MIL-E-4158.

3.2.3.2 Pressurization - The antenna shall be designed and constructed to hold internal pressure of three pounds per square inch so that when mounted vertically on a supporting structure water or other liquids cannot easily collect to an extent that could ultimately result in failure or deterioration of performance of the antenna. A relief valve, operable at a pressure of 10 pounds per square inch, shall be provided to facilitate purging the antenna.

3.2.3.3 Maintenance and Repair - The antenna shall be so constructed that maintenance and repairs can be made with a minimum of disturbance to other components. The radiating elements shall be readily demountable and interchangeable to provide ease of servicing and maintenance.

3.2.3.4 Mounting Brackets - The antenna shall have, at its base, a mounting flange with bolt holes to anchor the antenna firmly in position when it is installed aboard ship.

3.3 Performance Requirements

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:52Z
Check the source to verify that this is the current version before use.

3.3.1 Performance - The antenna shall be so manufactured that when mounted, in an unobstructed area, and sufficiently elevated above ground (or equivalent), the pattern of the antenna shall be such to provide essentially spatial coverage throughout a cylindrical volume.

3.3.2 Frequency - The antenna and feed system shall operate over the frequency range of 406 to 549 megacycles (mc).

3.3.3 Polarization - The antenna shall transmit essentially left-hand circularly polarized r-f energy, IRE Std.

3.3.4 Gain - The gain of the antenna shall be approximately five decibels (db) above an isotropic source.

3.3.5 Power Rating (R-F) - The antenna shall be capable of handling 1000 watts of r-f power over the frequency range 406 to 549 megacycles (mc).

3.3.6 Standing Wave Ratio, Voltage - The impedance at the input to the power divider shall match the 50-ohm impedance of the input coaxial cable so that the VSWR does not exceed the following:

Frequency	VSWR
406 to 425 mc	1.4:1
425 to 450 mc	1.6:1
450 to 500 mc	1.7:1
500 to 549 mc	1.8:1

3.4 Detail Requirements

3.4.1 Functional Description - The AT-781A/U shall have a set of radiators symmetrically disposed about a vertical supporting mast forming an array of three principal bays stacked equidistantly and excited to produce circularly polarized energy with omniazimuth and vertical plane coverage in the frequency range specified herein. A turnstile section of four quadrature unipoles shall be interposed between the second and third bays to improve overhead radiation coverage.

3.4.2 Turnstile Section - The turnstile section of the antenna shall improve overhead coverage. It shall consist of four quadrature unipoles phased by a network of RG-8/U cables which shall be housed in a turnstile cylinder mounted around the mast between the second and third dipole bays.

3.4.3 Dipole Bays - Each dipole bay shall consist of four broadband dipoles which shall be mounted symmetrically around the mast and inclined at an angle of approximately 45 degrees to the vertical axis of the mast.

3.4.4 Power Divider - The power divider, which shall be attached to the base of the AT-781A/U and which shall be an integral part of the antenna, shall be so constructed to feed one unit of power to each of the three dipole bays and 1/3 unit of power to the turnstile section.

3.4.5 Dimensions and Weight - The AT-781A/U shall have a maximum height of 80 inches and a maximum diameter, dipole to dipole, of 30 inches. The weight of the complete antenna shall not exceed 90 pounds.

3.4.6 R-F Connectors and Cable - The r-f input to the antenna shall be by means of the power divider which shall have a connector conforming to Specification MIL-C-71 which shall accept a 1-5/8 inch 50-ohm air-line coaxial cable or an adaptor for connecting a 1-5/8 inch input to RG-8/U or RG-17/U coaxial cable. The output from the power divider to the three bays and the turnstile section should be RG-8/U cables conforming to Specification MIL-C-17.

3.4.7 Nomenclature - Nameplate, nomenclature approval and assignment shall be granted by the Bureau of Aeronautics in compliance with the applicable requirements of Specification MIL-N-18307.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:52Z
Check the source to verify that this is the current version before use.

3.5 Workmanship - Workmanship shall be in accordance with Specification MIL-E-4158.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine compliance with all the applicable requirements.

(1) Preproduction Tests

(2) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. Data obtained by the contractor in conducting these tests shall be submitted to the Bureau of Aeronautics for review and approval prior to shipping the equipment to the specified destination for service approval tests. The Government Inspector and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the Bureau of Aeronautics.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the Bureau of Aeronautics, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include as a minimum the tests of 4.4.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the Bureau of Aeronautics shall engage the service of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the tests of 4.5 as a minimum.

4.4 Preproduction Tests - The preproduction model antenna shall be subjected to the tests in the specification as follows, and conducted at a laboratory designated by the Bureau of Aeronautics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:52Z
Check the source to verify that this is the current version before use.

4.4.1 Inclination - The antenna shall be subjected to inclination tests in accordance with MIL-T-17113.

4.4.2 Vibration - The antenna shall be subjected to vibration tests in accordance with MIL-T-17113.

4.4.3 Shock - The antenna shall be subjected to shock tests in accordance with MIL-T-17113.

4.5 Acceptance Tests - Each antenna produced shall be subjected to the acceptance tests detailed herein for acceptance before delivery. These tests are designed to insure that the complete antenna is in conformance with this specification.

4.5.1 Table of Tests - The tests to be performed shall be subdivided into two categories, (a) electrical, (b) mechanical.

4.5.1.1 Electrical Test No. 1 - Measurement of VSWR of complete antenna.

4.5.1.2 Mechanical Test No. 1 - Measurement of the pressurization and leakage rate of the antenna under three pounds of pressure. Test No. 2, perform mechanical and visual inspection of complete antenna.

4.5.2 Test Procedure - Unless specified otherwise, all tests shall be performed under the following conditions:

Atmospheric Conditions - The tests shall be performed at prevailing factory or laboratory atmospheric conditions.

4.5.2.1 Electrical Tests - The antenna shall be tested in accordance with an approved test procedure.

4.5.2.2 Mechanical Tests - The antenna shall be tested in accordance with an improved test procedure.

4.6 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the Bureau of Aeronautics for review and approval. The right is reserved by the Bureau of Aeronautics or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Bureau of Aeronautics will furnish the procedures to be used for testing.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the Bureau of Aeronautics.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Intended Use - The AT-781A/U antenna is intended for use with drone command transmitting and receiving equipment in the frequency range from 406 to 549 megacycles (mc). The antenna is a general purpose antenna, however, and can be used for other applications within the limits of its frequency range, power, and gain characteristics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:52Z
Check the source to verify that this is the current version before use.

6.2 This specification is not intended to be restricted with respect to the details of construction, except where such details are specified. Alternate forms of construction will be considered, provided that the contractor submits to the Bureau of Aeronautics a clear description for the points of difference of approval and provided that all performance requirements on this specification are met.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:52Z
Check the source to verify that this is the current version before use.